```
import numpy as np



A = np.array([
    [1.0, -0.81],
    [-0.09, 1.0]
])

b = np.array([-1.0, 89.9])


V = np.linalg.solve(A, b)
V_S0, V_S1 = V[0], V[1]


gamma = 0.9
Q_S0_Down = -1 + gamma * V_S1
Q_S0_Right = -1.0
Q_S1_Right = 100.0
Q_S1_Up = -1 + gamma * V_S0


V_S0_check = 0.9 * Q_S0_Down + 0.1 * Q_S0_Right
V_S1_check = 0.9 * Q_S1_Right + 0.1 * Q_S1_Up

print("=== ASSIGNMENT 1 RESULTS ===")
print(f"V*(S0)           : {V_S0:.2f}")
print(f"V*(S1)           : {V_S1:.2f}")
print(f"Q(S0, Down)      : {Q_S0_Down:.2f}")
print(f"Q(S0, Right)     : {Q_S0_Right:.2f}")
print(f"Q(S1, Right)     : {Q_S1_Right:.2f}")
print(f"Q(S1, Up)        : {Q_S1_Up:.2f}")
print("\n--- Verification ---")
print(f"V(S0) Calculated : {V_S0_check:.2f} (Matches V(S0): {np.isclose(V_S0, V_S0_check)})")
print(f"V(S1) Calculated : {V_S1_check:.2f} (Matches V(S1): {np.isclose(V_S1, V_S1_check)})")
```

```
=== ASSIGNMENT 1 RESULTS ===
V*(S0)           : 77.47
V*(S1)           : 96.87
Q(S0, Down)      : 86.18
Q(S0, Right)     : -1.00
Q(S1, Right)     : 100.00
Q(S1, Up)        : 68.72

--- Verification ---
V(S0) Calculated : 77.47 (Matches V(S0): True)
V(S1) Calculated : 96.87 (Matches V(S1): True)
```

```
import numpy as np
```

```
gamma = 0.9
theta = 1e-6


V = {'S0': 0.0, 'S1': 0.0}

iterations = 0


while True:
    V_old = V.copy()


    V['S0'] = 0.1 * (-1 + 0) + 0.9 * (-1 + gamma * V_old['S1'])
    V['S1'] = 0.9 * (100 + 0) + 0.1 * (-1 + gamma * V_old['S0'])

    iterations += 1

    delta = max(abs(V['S0'] - V_old['S0']), abs(V['S1'] - V_old['S1']))
    if delta < theta:
        break


Q_S0_Down = 0.1 * (-1) + 0.9 * (-1 + gamma * V['S1'])
Q_S0_Right = 1.0 * (-1)
Q_S1_Right = 0.9 * (100) + 0.1 * (-1 + gamma * V['S0'])
Q_S1_Up = 1.0 * (-1 + gamma * V['S0'])

print("=== ASSIGNMENT 2 RESULTS ===")
print(f"Number of evaluation iterations: {iterations}")
print(f"V*(S0)                         : {V['S0']:.2f}")
print(f"V*(S1)                         : {V['S1']:.2f}")
print(f"Q(S0, Down)                    : {Q_S0_Down:.2f}")
print(f"Q(S0, Right)                   : {Q_S0_Right:.2f}")
print(f"Q(S1, Right)                   : {Q_S1_Right:.2f}")
print(f"Q(S1, Up)                      : {Q_S1_Up:.2f}")
print("\n--- Policy Consistency Check ---")
print(f"V(S0) == Q(S0, Down)           : {np.isclose(V['S0'], Q_S0_Down)}")
print(f"V(S1) == Q(S1, Right)          : {np.isclose(V['S1'], Q_S1_Right)}")
```

```
=== ASSIGNMENT 2 RESULTS ===
Number of evaluation iterations: 15
V*(S0)                         : 77.47
V*(S1)                         : 96.87
Q(S0, Down)                    : 77.47
Q(S0, Right)                   : -1.00
Q(S1, Right)                   : 96.87
Q(S1, Up)                      : 68.72

--- Policy Consistency Check ---
V(S0) == Q(S0, Down)           : True
V(S1) == Q(S1, Right)          : True
```